



Superadmin Account Suspension & Deletion Implementation Plan

This plan covers account suspension and account deletion for both user-initiated and superadmin-initiated actions in Orviohub, with complete user journeys, user stories, Convex schemas, API endpoints, audit requirements, and NDPA 2023 compliance considerations. ^[1] ^[2] ^[3] ^[4] ^[5]

1. Key Definitions & Distinctions

Suspension vs Deletion

Aspect	Suspension	Deletion
Reversibility	Reversible (admin or user can restore)	Irreversible (after grace period)
Data retention	All data preserved, access blocked	Personal data erased/anonymized; business records retained per policy
Use case	Temporary (payment failure, policy violation, security concern)	Permanent (user request, workspace closure, compliance)
Access impact	User cannot sign in; workspace continues	User account removed; workspace ownership must be transferred first
NDPA compliance	Account deactivation ≠ data deletion ^[6]	Must erase personal data across systems (right to erasure) ^[3] ^[6]

User vs Superadmin Actions

Actor	Can Suspend	Can Delete
User (self)	✗ No (can only request deletion)	✓ Yes (own account, after workspace transfer/closure)
Workspace Owner	✓ Yes (members of their workspace only)	✗ No (cannot delete another user's personal account) ^[2]
Superadmin	✓ Yes (any user or workspace)	✓ Yes (any user or workspace, with audit)

2. User Journeys

2.1 User-Initiated Account Deletion

Entry point: `accounts.orviohub.com/profile/delete` ^[2]

Flow:

1. User navigates to deletion page

- System checks: user is authenticated, has no pending workspace ownership

2. Pre-deletion checklist displayed

- Workspaces owned → must transfer ownership or close workspace
- Active subscriptions → must cancel or let expire
- Pending invitations → auto-declined
- Consequences explained (data loss, cannot recover after grace period)

3. User confirms intent

- Re-authentication required (password or MFA)
- Explicit confirmation checkbox: "I understand this will permanently delete my account"

4. Deletion request created

- `accountDeletionRequests` record created with `scheduledDeletionAt = now + 7 days` (grace period)
- Confirmation email sent with cancellation link

5. Grace period (7 days)

- User can cancel deletion via email link or dashboard
- Account remains accessible but flagged

6. Scheduled deletion executes

- Personal data anonymized/erased per NDPA ^[4] ^[6]
- Business records (transactions, audit logs) preserved with anonymized user references
- Final confirmation email sent

User Stories:

- As a user, I want to delete my account so that my personal data is removed from Orviohub. ^[2]
- As a user, I want a grace period so that I can change my mind if I delete accidentally.
- As a user, I want clear warnings about what will be lost so that I can make an informed decision.

2.2 Superadmin-Initiated User Suspension

Entry point: `admin.orviohub.com/users/:userId`

Flow:

1. Superadmin searches for user

- Filter by email, name, workspace, status

2. Superadmin selects "Suspend User"

- Modal shows:
 - User's workspaces and roles
 - Active sessions
 - Impact: "User will be logged out immediately and cannot sign in"

3. Superadmin selects suspension reason

- Dropdown: `payment_failure`, `policy_violation`, `security_concern`, `fraud_suspected`, `other`
- Optional notes field (internal only)

4. Superadmin confirms

- Re-authentication recommended for destructive actions

5. Suspension executed

- `users.status = "suspended"`
- `suspendedAt`, `suspendedBy`, `suspensionReason` recorded
- All active sessions revoked
- User redirected to suspended page on next request
- Suspension email sent to user
- Audit event: `admin.user_suspended`

6. User impact

- Cannot sign in to any Orviohub product
- Workspace memberships preserved but inactive
- Billing continues unless workspace also suspended

User Stories:

- As a superadmin, I want to suspend a user immediately so that I can stop unauthorized access.
- As a superadmin, I want to record the suspension reason so that I have an audit trail.
- As a suspended user, I want to know why my account was suspended so that I can contact support.

2.3 Superadmin-Initiated User Deletion

Entry point: `admin.orviohub.com/users/:userId`

Flow:

1. Superadmin searches for user

- System shows: workspaces owned, memberships, subscriptions, last login

2. Superadmin selects "Delete User"

- **Critical checks:**
 - User owns workspaces? → Must transfer ownership first
 - User has active subscriptions? → Must cancel or transfer billing
 - User is the only superadmin? → **Block deletion** (must have ≥ 1 active superadmin)^[7]

3. Pre-deletion checklist

- Workspaces: transfer or close
- Subscriptions: cancel or reassign
- Data export available (optional, for compliance)

4. Superadmin confirms

- Re-authentication required
- Explicit confirmation: "This will permanently delete the user's account and personal data"

5. Deletion executed

- `accountDeletionRequests` created with `scheduledDeletionAt = now + 7 days`
- Or immediate deletion if `admin.force_delete = true` (bypass grace period, high-risk action)
- Personal data anonymized/erased
- Business records preserved with anonymized references
- Audit event: `admin.user_deleted`
- Notification to user (if email still valid)

User Stories:

- As a superadmin, I want to delete a user account so that I can remove fraudulent or abusive accounts.
- As a superadmin, I want to be prevented from deleting the last superadmin so that the platform remains manageable.
- As a superadmin, I want to force-delete immediately in emergencies so that I can respond to critical security incidents.

2.4 Superadmin-Initiated Workspace Suspension/Deletion

Entry point: `admin.orv.io/hub.com/workspaces/:workspaceId`

Flow (Suspension):

1. Superadmin selects workspace
 - Shows: owner, members, products, subscription status
2. Superadmin selects "Suspend Workspace"
 - Reason: `payment_failure`, `policy_violation`, `fraud`, `other`
 - Impact: "All members lose access; billing paused"
3. Confirmation
 - Re-authentication recommended
4. Suspension executed
 - `workspaces.status = "suspended"`
 - `suspendedAt`, `suspendedBy`, `suspensionReason` recorded
 - All product access revoked
 - All sessions invalidated
 - Workspace owner and members notified
 - Audit event: `admin.workspace_suspended`

Flow (Deletion):

1. Superadmin selects "Delete Workspace"
 - Checks: no active subscriptions or subscriptions cancelled
 - Data export available (optional)
2. Confirmation
 - Re-authentication required
 - Explicit confirmation
3. Deletion executed
 - `workspaces.status = "deleting"`
 - Grace period (7 days) or immediate if `admin.force_delete`
 - All business data erased per retention policy
 - Members notified
 - Audit event: `admin.workspace_deleted`

User Stories:

- As a superadmin, I want to suspend a workspace so that I can stop a business from using the platform due to non-payment or violations.

- As a superadmin, I want to delete a workspace so that I can remove fraudulent or closed businesses.
- As a workspace owner, I want to be notified if my workspace is suspended so that I can contact support.

3. Convex Schema Extensions

3.1 Users Table (Extended)

```
users: defineTable({
  // Existing fields
  email: v.string(),
  emailNormalized: v.string(),
  emailVerifiedAt: v.optional(v.number()),
  firstName: v.string(),
  lastName: v.string(),
  displayName: v.optional(v.string()),
  avatarUrl: v.optional(v.string()),
  phone: v.optional(v.string()),
  phoneVerifiedAt: v.optional(v.number()),
  country: v.string(),
  timezone: v.string(),
  status: v.string(), // "active", "suspended", "deleted"
  createdAt: v.number(),
  updatedAt: v.number(),
  lastLoginAt: v.optional(v.number()),

  // NEW: Suspension fields
  suspendedAt: v.optional(v.number()),
  suspendedBy: v.optional(v.id("users")), // superadmin who suspended
  suspensionReason: v.optional(v.string()), // "payment_failure", "policy_violation"
  suspensionNotes: v.optional(v.string()), // internal notes

  // NEW: Deletion tracking
  deletionRequestedAt: v.optional(v.number()),
  deletionScheduledAt: v.optional(v.number()),
  deletedAt: v.optional(v.number()),
  deletedBy: v.optional(v.id("users")), // null if self-deletion
})
.index("by_email", ["emailNormalized"])
.index("by_status", ["status"])
.index("by_status_suspended", ["status", "suspendedAt"])
```

3.2 Account Deletion Requests Table

```
accountDeletionRequests: defineTable({
  userId: v.id("users"),
  status: v.string(), // "pending", "cancelled", "completed"
  reason: v.optional(v.string()), // user-provided reason
  requestedBy: v.id("users"), // user or superadmin
  requestedAt: v.number(),
```

```

scheduledDeletionAt: v.number(), // grace period end
cancelledAt: v.optional(v.number()),
completedAt: v.optional(v.number()),
deletionTokenHash: v.optional(v.string()), // for email cancellation link
adminForceDelete: v.optional(v.boolean()), // true if bypassed grace period
}))
.index("by_user", ["userId"])
.index("by_status", ["status"])
.index("by_scheduled_deletion", ["scheduledDeletionAt"])

```

3.3 Workspaces Table (Extended)

```

workspaces: defineTable({
  // Existing fields
  name: v.string(),
  slug: v.string(),
  type: v.string(),
  ownerId: v.id("users"),
  country: v.string(),
  timezone: v.string(),
  currency: v.string(),
  status: v.string(), // "active", "suspended", "archived", "deleting", "deleted"
  createdAt: v.number(),
  updatedAt: v.number(),

  // NEW: Suspension/deletion tracking
  suspendedAt: v.optional(v.number()),
  suspendedBy: v.optional(v.id("users")),
  suspensionReason: v.optional(v.string()),
  deletionRequestedAt: v.optional(v.number()),
  deletionScheduledAt: v.optional(v.number()),
  deletedAt: v.optional(v.number()),
  deletedBy: v.optional(v.id("users")),
}))
.index("by_owner", ["ownerId"])
.index("by_status", ["status"])
.index("by_status_suspended", ["status", "suspendedAt"])

```

3.4 Audit Logs Table (Extended)

```

auditLogs: defineTable({
  actorUserId: v.optional(v.id("users")),
  targetUserId: v.optional(v.id("users")),
  targetWorkspaceId: v.optional(v.id("workspaces")),
  eventType: v.string(),
  entityType: v.string(), // "user", "workspace", "subscription"
  entityId: v.string(),
  severity: v.string(), // "low", "medium", "high", "critical"
  metadata: v.any(), // { reason, adminForceDelete, etc. }
  ipAddress: v.optional(v.string()),
  userAgent: v.optional(v.string()),
  requestId: v.optional(v.string()),
  createdAt: v.number(),

```

```
  })
  .index("by_actor", ["actorUserId"])
  .index("by_target_user", ["targetUserId"])
  .index("by_target_workspace", ["targetWorkspaceId"])
  .index("by_event_type", ["eventType"])
  .index("by_created", ["createdAt"])
```

4. API Endpoints

4.1 User-Initiated Deletion

```
POST    /v1/users/me/deletion/request
DELETE  /v1/users/me/deletion/cancel
GET     /v1/users/me/deletion/status
```

Request (POST /v1/users/me/deletion/request):

```
{
  "reason": "no_longer_needed",
  "confirmPassword": "user_password"
}
```

Response:

```
{
  "deletionRequestId": "del_123",
  "scheduledDeletionAt": 1726000000000,
  "gracePeriodDays": 7,
  "message": "Account deletion scheduled. Check email for cancellation link."
}
```

4.2 Superadmin User Suspension

```
POST    /admin/v1/users/:userId/suspend
POST    /admin/v1/users/:userId/restore
GET     /admin/v1/users/:userId/suspension-history
```

Request (POST /admin/v1/users/:userId/suspend):

```
{
  "reason": "policy_violation",
  "notes": "User violated terms of service - spam activity",
  "revokeAllSessions": true
}
```

4.3 Superadmin User Deletion

```
POST    /admin/v1/users/:userId/delete
POST    /admin/v1/users/:userId/delete/force  // bypass grace period
```

Request (POST /admin/v1/users/:userId/delete):

```
{
  "reason": "fraud",
  "notes": "Fraudulent account - multiple payment chargebacks",
  "transferWorkspaceOwnership": true,
  "newOwnerId": "user_456",
  "cancelSubscriptions": true,
  "adminForceDelete": false
}
```

4.4 Superadmin Workspace Suspension/Deletion

```
POST    /admin/v1/workspaces/:workspaceId/suspend
POST    /admin/v1/workspaces/:workspaceId/restore
POST    /admin/v1/workspaces/:workspaceId/delete
POST    /admin/v1/workspaces/:workspaceId/delete/force
```

5. Business Logic & Validation Rules

5.1 Suspension Rules

User Suspension:

- ✓ Superadmin can suspend any non-superadmin user
- ✓ Superadmin can suspend another superadmin only if ≥ 2 active superadmins exist^[7]
- ✗ User cannot suspend themselves
- ✓ Suspended user: all sessions revoked, cannot sign in, workspace memberships preserved

Workspace Suspension:

- ✓ Superadmin can suspend any workspace
- ✓ All members lose access immediately
- ✓ Billing paused (subscription status $\rightarrow$ past_due or suspended)
- ✓ Data preserved for restoration

5.2 Deletion Rules

User Deletion (Self-Initiated):

- ✓ User can request deletion only if:
 - No workspaces owned (or ownership transferred)
 - No active subscriptions (or cancelled)
 - Not the only superadmin
- ✓ 7-day grace period (configurable)
- ✓ User can cancel via dashboard or email link
- ✓ After grace period: personal data erased, business records anonymized

User Deletion (Superadmin-Initiated):

- ✓ Superadmin can delete any user **except**:
 - Themselves (cannot delete own account via admin panel)
 - The only remaining superadmin^[7]
- ✓ Can bypass grace period with `adminForceDelete: true` (high-risk, requires audit)
- ✓ Must handle workspace ownership transfer before deletion

Workspace Deletion:

- ✓ Only workspace owner or superadmin can request
- ✓ Must cancel subscriptions or transfer billing
- ✓ Must notify all members
- ✓ Business data erased per retention policy (NDPA: keep financial records 6-7 years)^[4]

6. Notification & Email Templates

6.1 User-Initiated Deletion

Email 1: Deletion Request Confirmation

Subject: Your Orviohub account deletion request

Hi [Name],

We received your request to delete your Orviohub account.

Your account is scheduled for deletion on [Date - 7 days from now].

If you change your mind, you can cancel here: [Cancel Link]

What will be deleted:

- Your personal profile
- Your login credentials

- Your notification preferences

What will be preserved:

- Business records (transactions, inventory) with your name anonymized
- Audit logs for compliance

Questions? Reply to this email.

— Orviohub Team

Email 2: Final Deletion Confirmation

Subject: Your Orviohub account has been deleted

Hi [Name],

Your Orviohub account has been permanently deleted.

If you have questions, contact support at support@orviohub.com.

— Orviohub Team

6.2 Superadmin-Initiated Suspension

Email to User:

Subject: Your Orviohub account has been suspended

Hi [Name],

Your Orviohub account has been suspended by our admin team.

Reason: [Suspension Reason]

You cannot sign in to Orviohub until this suspension is lifted.

If you believe this is a mistake, contact support at support@orviohub.com.

— Orviohub Team

6.3 Superadmin-Initiated Deletion

Email to User (if email still valid):

Subject: Your Orviohub account has been deleted

Hi [Name],

Your Orviohub account has been deleted by our admin team.

Reason: [Deletion Reason]

If you have questions, contact support at support@orviohub.com.

– Orviohub Team

7. Audit Log Events

Event	Severity	Metadata Fields
user.deletion_requested	medium	reason, gracePeriodDays
user.deletion_cancelled	medium	cancelledBy
user.deletion_completed	high	deletedBy, anonymizedRecords
admin.user_suspended	high	reason, notes, adminUserId
admin.user_restored	high	adminUserId
admin.user_deleted	critical	reason, adminForceDelete, adminUserId
admin.workspace_suspended	high	reason, adminUserId
admin.workspace_deleted	critical	reason, adminForceDelete, adminUserId

8. NDPA 2023 Compliance Checklist

- ✓ **Right to erasure:** Users can request deletion; personal data erased within grace period^[5]
^[6]
- ✓ **Data retention:** Business records (transactions, audit logs) retained per legal requirements (6-7 years for financial data)^[4]
- ✓ **Audit trail:** All admin actions logged with actor, timestamp, reason^[6]
- ✓ **Grace period:** 7-day cooling-off period for user-initiated deletion^[8]
- ✓ **Anonymization:** Deleted user references in business records anonymized (e.g., "Deleted User #123")
- ✓ **Notification:** Users notified of suspension/deletion with reason and support contact
- ✓ **Data export:** Users can request data export before deletion (GDPR/NDPA right to portability)^[2]

9. Implementation Phases

Phase 1: Schema & Core Logic (Week 1)

- Add suspension/deletion fields to users, workspaces, accountDeletionRequests
- Implement indexes for efficient queries
- Create audit log event types

Phase 2: User-Initiated Deletion (Week 2)

- Build `accounts.orv.iohub.com/profile/delete` page
- Implement deletion request API with grace period
- Send confirmation emails with cancellation links
- Create background job for scheduled deletions

Phase 3: Superadmin Suspension (Week 3)

- Build `admin.orv.iohub.com/users` list with suspension action
- Implement suspension API with reason tracking
- Revoke all sessions on suspension
- Send suspension notification emails

Phase 4: Superadmin Deletion (Week 4)

- Build deletion confirmation modal with pre-checks
- Implement workspace ownership transfer flow
- Add `adminForceDelete` bypass for emergencies
- Log all admin deletion actions with high-severity audit

Phase 5: Workspace Suspension/Deletion (Week 5)

- Extend suspension/deletion logic to workspaces
- Handle subscription cancellation/transfer
- Notify all workspace members
- Implement data archival before deletion

Phase 6: Testing & Hardening (Week 6)

- Test edge cases: last superadmin, workspace owner deletion, concurrent requests
- Verify NDPA compliance (data erasure, audit logs, retention)
- Load test with large user/workspace datasets
- Security review (privilege escalation, IDOR prevention) ^[9]

10. Edge Cases & Error Handling

Scenario	Handling
User is only superadmin	Block deletion; show error: "Cannot delete the only superadmin. Add another superadmin first." [2]
User owns workspaces	Require ownership transfer before deletion; provide transfer flow
User has active subscription	Require cancellation or transfer to another user
Concurrent deletion requests	First request wins; subsequent requests return existing <code>deletionRequestId</code>
User signs in during grace period	Show banner: "Your account is scheduled for deletion on [date]. [Cancel]"
Email bounce on deletion notification	Log failure; proceed with deletion after grace period (email delivery not blocking)
Superadmin deletes themselves	Block via API validation; must use different admin account
Workspace has no owner after deletion	Prevent user deletion until ownership transferred; transactional rollback

11. Security Considerations

- **Re-authentication:** Require password/MFA for deletion and suspension actions [2]
- **Rate limiting:** Prevent abuse (max 3 deletion requests per user per month)
- **IDOR prevention:** Validate workspace/user ownership on every request [9]
- **Audit immutability:** Audit logs cannot be edited/deleted (append-only)
- **Session revocation:** Immediately invalidate all sessions on suspension/deletion
- **Privilege escalation protection:** Prevent admins from suspending/deleting superadmins without checks [9]

This plan provides a complete, NDPA-compliant implementation for account suspension and deletion in Orvionhub. Ready to proceed with specific implementation (e.g., "implement Phase 1 schemas" or "build the user deletion flow")?

✱

1. docs/authentication and onbaording plans.pdf
2. docs/User Personal Profile Management Plan.pdf
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